MIL-C-48162B (AR)
14 August 1987
SUPERCEDING
MIL-C-48162A
3 March 1980

MILITARY SPECIFICATION

CARTRIDGE, IGNITION, M702
LOADING, ASSEMBLING AND PACKING

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and provides quality assurance provisions for the loading, assembling and packing of the M702 Ignition Cartridge for the 60MM Light Weight Company Mortar System.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-223 - Powder, Black
MIL-P-46425 - Primer, Percussion, M35, Loading
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament,Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT DRAWINGS (See 6.4)

PRODUCT AND PACKAGING DRAWINGS

9280553 - Cartridge, Ignition, M702 Loading Assembly
9287826 - Packing and Marking for Box, Fiberboard, Packing, for Cartridge Ignition, M702

INSPECTION EQUIPMENT DRAWINGS

9278622 - Envelope Drawing for Leak Testing of Ignition Cartridges

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

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3.2 Assembly. The ignition cartridge assembly and components shall comply with all requirements specified on Drawing (dwg) 9280553 and with all requirements specified in applicable specifications, associated drawings and standards.

3.3 Moisture content of black powder. The moisture content of the black powder used in loading the ignition cartridge, at the time of loading, shall not exceed 0.50 percent. Testing shall be as specified in 4.5.1.

3.4 Sensitivity.

3.4.1 Nonfunctioning. The primer shall not function under an applied energy value of 8.85 inch ounces minimum (min.) when tested as specified in 4.5.2.1.

3.4.2 Functioning. The primer shall function under an applied energy value of 82.60 inch ounces maximum (max.) when tested as specified in 4.5.2.2.

3.4.3 Safety drop test. The primer shall not function as a result of the five foot cartridge drop specified in 4.5.2.3.

3.5 Leak test. The loaded cartridges shall satisfactorily pass the leak test specified in 4.5.3.

3.6 Velocity, functioning and security (proving ground test). The ignition cartridges shall comply with the following requirements (see 4.5.5).

3.6.1 Misfires. The ignition cartridges shall function on the first attempt, without any audible hangfire.

3.6.2 Metal part security. There shall be no evidence of rupture or metal parts separation in the gun bore or in flight.

3.6.3 Velocity and functioning (proving ground). The cartridge assemblies shall function and shall impart a corrected muzzle velocity of not more than 219 feet per second (fps) and not less than 199 fps. In addition, the velocity standard deviation of the lot shall be not greater than 4.0 fps.

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3.7 Propellant charge uniformity. The M9 propellant lots shall achieve a velocity of 210 fps (see 4.5.6). This test shall be required when a lapse of two years or more occurs between propellant acceptance and LAP initiation.

3.8 Static pressure. The average internal ignition cartridge pressure plus 4 standard deviations shall not exceed 22,000 psi (see 4.5.7).

3.9 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein. (See 4.3 and 6.2)

3.10 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the components in the quantities indicated in Table I.

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48162B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER SEE BELOW
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Tube Assembly (dwg. 12561285)				
	Examination for defects	15		3.2	4.4.2.1
	Pellet (dwg. 12561290)				
	Examination for defects	15		3.2	4.4.2.2
	Moisture content	10		3.3	4.5.1
	Flash Tube Loading Assembly (dwg. 9280529)				
	Examination for defects	15		3.2	4.4.2.3
	Cartridge, Ignition, M702, Loading Assembly (dwg. 9280553)				
	Examination for defects	15		3.2	4.4.2.4
	Weight of propellant	15		3.2	4.4.3.1/4.5.4
	Leak test	15		3.5	4.5.3
	Velocity, functioning and security	50		3.6	4.4.3.3/4.5.5
	Sensitivity functioning	15		3.4	4.4.3.2.1/4.5.2.1
	Sensitivity non-functioning	15		3.4	4.4.3.2.2/4.5.2.2
	Static pressure	100		3.8	4.4.3.5/4.5.7

NOTES:

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Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of ignition cartridges shall contain:

a. Metal parts of one lot interfix from one manufacturer.

b. Black powder from not more than one lot from one manufacturer.

c. Percussion elements from not more than one lot interfix from one manufacturer.

d. M9 propellant from not more than one lot from one manufacturer.

e. Tubing from not more than one manufacturer or manufacturer part number.

f. Adhesive sealant from one manufacturer.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48162B(AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.1	Tube Assembly		1 OF 1		12561285
					NEXT HIGHER ASSEMBLY
					9280553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter, max.		0.40%	3.2	Gage
102	Tubing missing, loose, wrinkled, separating or inadequately covering external surface		0.40%	3.2	Visual/Manual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48162B (AR)

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER 12561290
4.4.2.2	Pellet				NEXT HIGHER ASSEMBLY 9280529
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight of pellet	25*	0.40%	3.1	Balance
102	Moisture content			3.3	4.5.1
103	Outer diameter		0.40%	3,2	Gage
104	Inner diameter		0.40%	3.2	Gage
105	Length		0.40%	3.2	Gage
Minor	None defined				

NOTES:
*Test shall be performed on material selected from each eight hour production.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48162B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9280529
4.4.2.3	Flash Tube Loading Assembly				NEXT HIGHER ASSEMBLY 9280553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Tubing missing, loose or does not cover all flash holes		100%	3.2	Visual/Manual
Major					
101	Pin missing		100%	3.2	Visual
102	Pellet not flush or below flush		0.40%	3.2	Gage
103	Adhesive on pellet surface or perforation		100%	3.2	Visual
104	Pellet missing		100%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3:10	Visual

NOTES:

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Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48162B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9280553
4.4.2.4	Cartridge, Ignition Prior to Assembly of Head				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Percussion primer missing		100%	3.2	Visual
102	Percussion primer inverted		100%	3.2	Visual
103	Lacquer seal missing or inadequate		0.40%	3.2	Visual
104	Lacquer on primer face		0.40%	3.2	Visual
105	Percussion primer above flush (max.) of body		0.40%	3.2	Gage
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48162B (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.5	Cartridge, Ignition Prior to Assembly of Cap		1 OF 1	9280553
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Flash tube loading assembly missing		100%	3.2	Visual
Major					
101	Propellant weight (before loading)		100%	3.2	4.4.3.1.1/Scale
102	Propellant missing		100%	3.2	Visual
103	Body stakes missing		0.40%	3.2	Visual
104	Environmental conditions inadequate	/a		3.2	Gage
Minor					
201	Foreign matter in propellant charge		0.65%	3.10	Visual

NOTES: a. Temperature and humidity shall be checked at the beginning and end of each shift. If conditions are not proper at either time, the quantity of product represented by the conditioning shall be rejected for further processing.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Cartridge, Ignition	1 OF 2	9280553
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Firing plug above flush with head surface		100%	3.2	Gage
2	Sensitivity nonfunctioning	200*		3.4.1	4.4.3.2.1/4.5.2.1
3	Sensitivity safety drop test	25*		3.4.3	4.4.3.2.3/4.5.2.3
Special					
A	Firing plug not loose after assembly		100%	3.2	Manual
B	Weight of propellant (after loading)	**		3.2	4.4.3.1.2/4.5.4
Major					
101	RTV sealant missing, inadequate or excessive		0.40%	3.2	Visual
102	Length, maximum (from cap to seat of head)		0.40%	3.2	Gage
103	Weight of propellant (after loading)	**		3.2	4.4.3.1.2/4.5.4
104	Leak test		100%	3.5	4.5.3
105	Velocity, functioning and security (proving ground)	***		3.6	4.4.3.3/4.5.5
106	Sensitivity functioning	500		3.4.2	4.4.3.2.2/4.5.2.2
107	Firing plug missing or inverted		0.40%	3.2	Visual

NOTES: *Tests performed after assembling primer, head, firing plug and body.
**See 4.4.3.1.2 for defect description. Same sample used for both inspections.
***Sample size per 4.4.3.3.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48162B (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.6	Cartridge, Ignition (Cont)				9280553
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
108	True position		0.40%	3.2	Gage
109	Marking missing, misleading or unidentifiable		0.40%	3.2	Visual
110	Environmental conditions inadequate	/a		3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

a. Temperature and humidity shall be checked at the beginning and end of each shift. If conditions are not proper at either time, the quantity of product represented by the conditioning shall be rejected for further processing.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48162B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287827
4.4.2.7	Tray Assembly for Cartridge, Ignition, M702				NEXT HIGHER ASSEMBLY 9287826
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Tray broken, split or incomplete		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Box, Fiberboard, Before Packing	1 OF 1	9287826
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside dimensions incorrect		0.40%	3.2	Gage
102	Broken, split or missing components		0.40%	3.2	Visual
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48162B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Packed Fiberboard Box, Before Closing		9287826
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cartridges missing		0.40%	3.2	Visual
102	Improperly packed		0.40%	3.2	Visual
103	Fillers missing		0.40%	3.2	Visual
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287826
4.4.2.10	Packed Fiberboard Box, After Closing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Improperly sealed		0.40%	3.2	Visual
Minor 201	Contents loose		0.65%	3.2	Manual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Weight of propellant

4.4.3.1.1 Before loading. (See dwg. 9280553). The propellant charge shall be weighed or volumetrically dispensed into a tared container. Weight limits for weighing should be established within the applicable requirements, depending upon the accuracy and precision of the weighing system to assure compliance with the charge weight requirements. Any charge of bulk propellant failing to meet the applicable requirements shall be classed defective and removed from the lot.

4.4.3.1.2 After loading. Weight 5 grains or less above assessed maximum weight - Major Defect: Weight more than 5 grains above assessed maximum weight - Special Defect: Weight 2 grains or less below assessed minimum weight - Major Defect: Weight more than 2 grains below assessed minimum weight - Special Defect. 315 sample cartridge assemblies shall be randomly selected from each lot for this test. If four or more samples exceed the assessed max weight by 5 or less grains or if any sample exceeds the assessed max weight by more than 5 grains, the lot shall be rejected. If four or more samples are under the assessed minimum weight by 2 grains or less or if any sample is under the assessed minimum weight by more than 2 grains, the lot shall be rejected. In addition, if the total number of defective samples are five or more, the lot shall be rejected. Testing shall be as specified in 4.5.4.

4.4.3.1.3 Automated method. The contractor may elect to perform this operation with automated equipment and mandatory process controls. In such cases, process control plans in accordance with Data Item Description DI-P-1604 must be submitted to the Technical Agency for approval of the automated operation.

4.4.3.2 Sensitivity

4.4.3.2.1 Nonfunctioning. Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, 200 primed heads shall be randomly selected from each lot for this test. If one or more cartridges fail to comply with the specified requirement, the lot shall be rejected. The test shall be performed as specified in 4.5.2.1.

4.4.3.2.1.1 Nonfunctioning. After three consecutive lots have met the criteria of 4.4.3.2.1, the sampling plan for this test shall be in accordance with MIL-STD-105, Inspection Level II Code Letter K and

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AQL of 0.10 percent. If a lot is rejected, revert to sampling plan per 4.4.3.2.1. The test shall be performed as specified in 4.5.2.1.

4.4.3.2.2 Functioning. Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, 500 primed heads shall be randomly selected from each lot for this test. If 4 or more primers fail to comply with the specified requirement, the lot shall be rejected. If 2 but less than 4 defectives are found a second sample of 500 primed heads shall be tested. If the combined number of defectives in both the first and second sample is 5 or more, the lot shall be rejected. Any primed heads in the randomly selected sample missing a component or with an incorrectly assembled component which will result in an end item misfire shall be considered defective and shall be included in the rejection criteria. The test shall be performed as specified in 4.5.2.2.

4.4.3.2.2.1 Functioning. After three consecutive lots have met the criteria of 4.4.3.2.2, a sample of 315 primed heads shall be randomly selected from each lot for this test. If 3 or more primers fail to comply with the specified requirement, the lot shall be rejected. If 1 but less than 3 defectives are found a second sample of 315 primed heads shall be tested. If the combined number of defectives in both the first and second sample is 4 or more the lot shall be rejected. Any primed heads in the randomly selected sample missing a component or with an incorrectly assembled component which will result in an end item misfire shall be considered defective and shall be included in the rejection criteria. If a lot is rejected, revert to sampling plan per 4.4.3.2.2. The test shall be performed as specified in 4.5.2.2.

4.4.3.2.3 Safety drop test. Twenty-five test assemblies consisting of the primed body, head, firing plug, flash tube/pellet assembly, tube assembly and cap (assembled without M9 propellant in accordance with dwg. 9280553) shall be randomly selected for this test. If one or more of the test assemblies function when tested as specified in 4.5.2.3, the primer lot shall be rejected. This test shall be performed only upon initial use of each primer lot.

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4.4.3.3 Velocity, functioning and security.

4.4.3.3.1 Velocity, functioning and security. Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code letter M and AQL of 1.0 percent. In addition, if the velocity standard deviation of the sample exceeds the lot requirement, the lot shall be rejected. If any round fails to function for reasons attributable to ignition cartridge failure or if a metal parts separation occurs, the lot shall be rejected. This test shall be performed as specified in 4.5.5.

4.4.3.3.2 Velocity, functioning and security. After three consecutive lots have met the criteria of 4.4.3.3.1, the sampling plan for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, code letter I, using an AQL of 1.0 percent. In addition, if the velocity standard deviation of the sample exceeds the applicable lot requirement the lot shall be rejected. If any round fails to function for reasons attributable to ignition cartridge failure, or if a metal parts separation occurs, the lot shall be rejected. If a lot is rejected, revert to sampling plan per 4.4.3.3.1. This test shall be performed as specified in 4.5.5.

4.4.3.4 Check test for deterioration of primer, percussion M35. If the total elapsed time between original acceptance of the M35 primer lot and the assembly of that lot into the ignition cartridge exceeds two years, or if the M35 primers have been subjected to adverse conditions for any period of time since previous test, the M35 primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-P-46425 immediately before the M35 primer lot is assembled into the ignition cartridges.

4.4.3.5 Static pressure. The M9 propellant lot shall be tested as specified in 4.5.7. One hundred (100) ignition cartridges shall be loaded at the assessed charge weight with M9 propellant randomly selected from the propellant lot. If the average ignition cartridge pressure exceeds the specified requirement the representative lot of M9 propellant shall be withheld and referred to the engineering agency.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with 4.4.2.1. The contractor shall

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submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Moisture content of black powder. The moisture content of black powder shall be determined in accordance with the procedure specified in MIL-P-223. The samples shall be randomly selected from each eight hours production. Any failure to comply with the applicable requirement shall be cause for rejection of the eight hours production represented by the sample. If loading has begun, ignition cartridges loaded with the non-conforming black powder shall be rejected.

4.5.2 Sensitivity

4.5.2.1 Nonfunctioning. The primer head shall be assembled to a suitable fixture. The firing pin of the test fixture shall be adjusted so that impact will be on the center of the firing plug of the primer. A weight of 2.95 pounds, minimum, shall be dropped on to the firing pin from a height of 3/16 inch, minimum, as if to function the primer. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight. Any primer which functions shall be classed defective. Primers used in this test shall not be returned to the lot.

4.5.2.2 Functioning. The primer head shall be assembled to an appropriate test fixture. The firing pin of the test fixture shall be adjusted so that impact will be 0.05 inch off center of the firing plug of the primer. A weight of 2.95 pound, maximum, shall be dropped on the firing pin to function the primer from a height of 1 3/4 inches, maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight. Any primer that fails to function shall be classed defective.

4.5.2.3 Safety drop test. The test assemblies shall be assembled into inert-loaded, standard 60mm cartridges. Each cartridge shall be dropped twice from a height of five feet on an anvil top surface. For the first drop, the cartridge shall be in the base (fin) down orientation. For the second drop, the cartridge shall be in the nose (fuze) down orientation. The drop height shall be measured from the lowest point of the cartridge orientation to the top of the anvil surface. Any test assembly which functions as a

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result of the drop shall be classed defective. All questionable test assemblies shall be downloaded to verify primer/black powder pellet functioning or non-functioning. All components used for this test shall not be returned to their respective lots.

4.5.3 Leak test. After the ignition cartridges have been assembled and the RTV is cured, the ignition cartridges will be kept in uncontrolled (temperature) storage for 72 hours minimum prior to performing the leak test. The ignition cartridges shall be tested 100% in accordance with drawing 9278622. Any ignition cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot.

4.5.4 Weight of propellant. The sample cartridge assemblies shall be cut open using any method that will not generate chips and all propellant removed and placed on the pan of a precision analytical balance. The weight of propellant from each sample shall be determined to the nearest 0.01 grain.

4.5.5 Velocity, functioning and security (proving ground test). The test shall be performed at a Government Proving Ground in accordance with the applicable USA TECOM Acceptance Test Procedure.

The sample ignition cartridges shall be assembled with a standard finned projectile (3.75 pounds (lbs)) for which they were manufactured. The complete assembly, less propelling charges, shall be fired from a 60MM LWCM ballistic tube at an elevation of forty-five degrees.

Reference rounds shall be fired without propelling charges for the purpose of velocity correction. At least one reference round should be fired for every three test rounds.

Individual velocities shall be recorded on the firing record for both reference and test rounds. Velocity standard deviations and arithmetic averages shall be calculated and recorded. Temperature of the day and time of firing shall also be recorded. In the event of a misfire, the misfired ignition cartridge shall not be disassembled by the proving ground but shall be forwarded to the responsible technical activity. Individual chamber pressures shall be obtained for each round firing.

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4.5.5.1 Refiring. If for any reason the proving ground considers that conditions have detrimentally affected the test results, additional cartridges as required shall be tested.

4.5.6 Propellant charge uniformity. (To be conducted at a U.S. Government Proving Ground). Ten test ignition cartridges shall be loaded at charge weight with a tolerance of plus or minus 0.5 grains. The ignition cartridges shall be assembled to 60MM production rounds. Rounds shall be inert loaded to achieve a total weight of 3.75 pounds plus or minus .02 pounds. The rounds shall be conditioned at 70°F (+2.5°F) for 24 hours prior to firing and shall be maintained at this temperature until fired. The rounds shall be fired from a 60MM LWCMS ballistic tube at an elevation of forty-five degrees. Calibration rounds shall be fired alternately for velocity correction. The mean corrected velocities of the group shall be compared to the established charge weight velocity. If the corrected velocity of the test rounds exceeds two percent of the expected muzzle velocity, use of the propellant lot shall be referred. Test results shall be forwarded to the technical agency for evaluation (see 6.5).

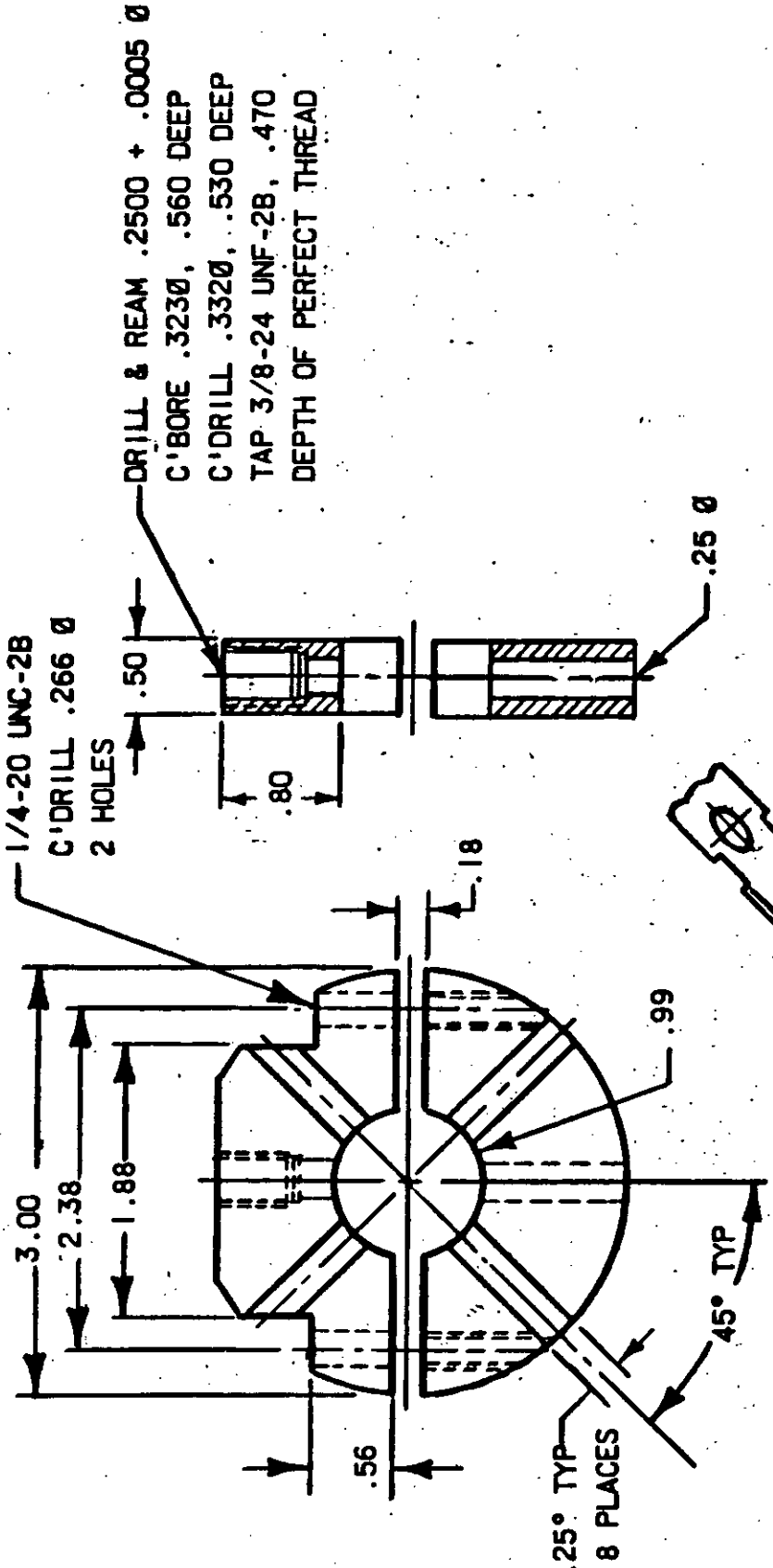
4.5.7 Static pressure test. The ignition cartridges shall be threaded into the M27 Fin and firmly seated. A 3/32 inch hole shall be drilled in the ignition cartridge tube through one flash hole in the seventh row of flash holes in the fin boom, counting from the ignition cartridge end. A pressure transducer, mounted to a collar (see SKETCH 1), shall be attached to the fin boom, positioning the collar so that the transducer will be directly over the flash hole drilled in the ignition cartridge tube. The collar shall be positioned carefully to prevent restriction of venting from other flash holes in the boom. The assembly shall be set up as shown in SKETCH 2 and the transducer connected to a recorder. The ignition cartridge shall then be functioned and pressure/time tracer recorded.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be as specified in Dwg 9287826.

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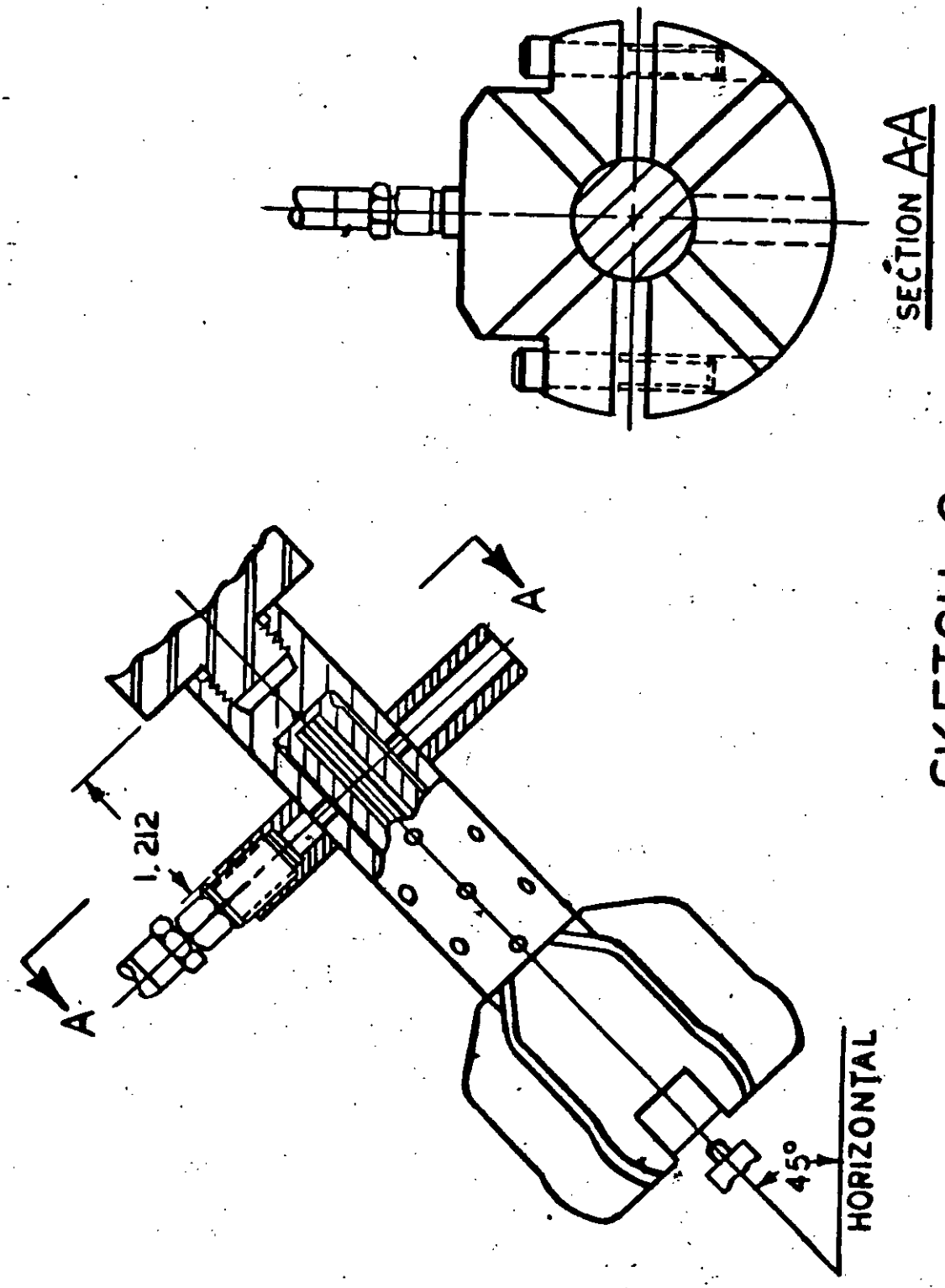
COLLAR



SKETCH I

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M702 STATIC IGNITION CARTRIDGE TEST SETUP



SKETCH 2

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6. NOTES

6.1 Intended use. The ignition cartridges covered by this specification are intended for use on cartridges for the 60MM Lightweight Company Mortar System (LWCMS).

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract: Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or ARRADCOM. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Submission of test data. Formal firing reports containing velocity test data and lot acceptability shall be sent to Commander, US Army Armament Research and Development Center, ATTN: AMSMC-QAT-B, SMCAR-FSA-IM, Picatinny Arsenal, NJ 07806-5000.

6.6 Check test for possible deterioration of cartridge, ignition. The following test is designed to prevent the loading of major ammunition items with ignition cartridges which have deteriorated. The test shall be performed on each lot of ignition cartridges which has been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the ignition cartridge is assembled does not require that the check test be performed, the contracting officer shall insert a provision for the performance of the check test in the contract. The contracting officer shall arrange to have the contractor reimbursed for any expense incurred in the performance of the check test for possible deterioration.

6.6.1 Velocity, functioning and security. A sample from each lot shall be selected in accordance with MIL-STD-414, Section B, Table B-3, Code letter M and AQL of 1.0 percent. In addition, if the velocity standard deviation of the sample exceeds

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the lot requirement, the shall be rejected. If any round fails to function for reasons attributable to ignition cartridge failure, or if a metal parts separation occurs, the lot shall be rejected. This test shall be performed as specified in 4.5.5.

TEST	SAMPLE SIZE	REQUIREMENTS
First Article		
Velocity	50 - HE Loaded M720 Cartridge Less Propelling Charges 16 - M720 Reference Calibration Cartridges	3.6
First Three Production Lots		
Velocity	50 - HE Loaded, M720 Cartridges Less Propelling Charges 16 - M720 Reference Calibration Cartridges	3.6

6.7 Data Requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.	Option
4.4.3.1.3	Process Control Plan	DI-P-1604	--

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.8 Subject term (key word) listing.

Cartridge, ignition
Functioning
Leak test
Pellet
Security
Sensitivity
Velocity

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issues due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project: 1310-A380)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-48162B	CARTRIDGE, IGNITION, M702 LOADING, ASSEMBLING AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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